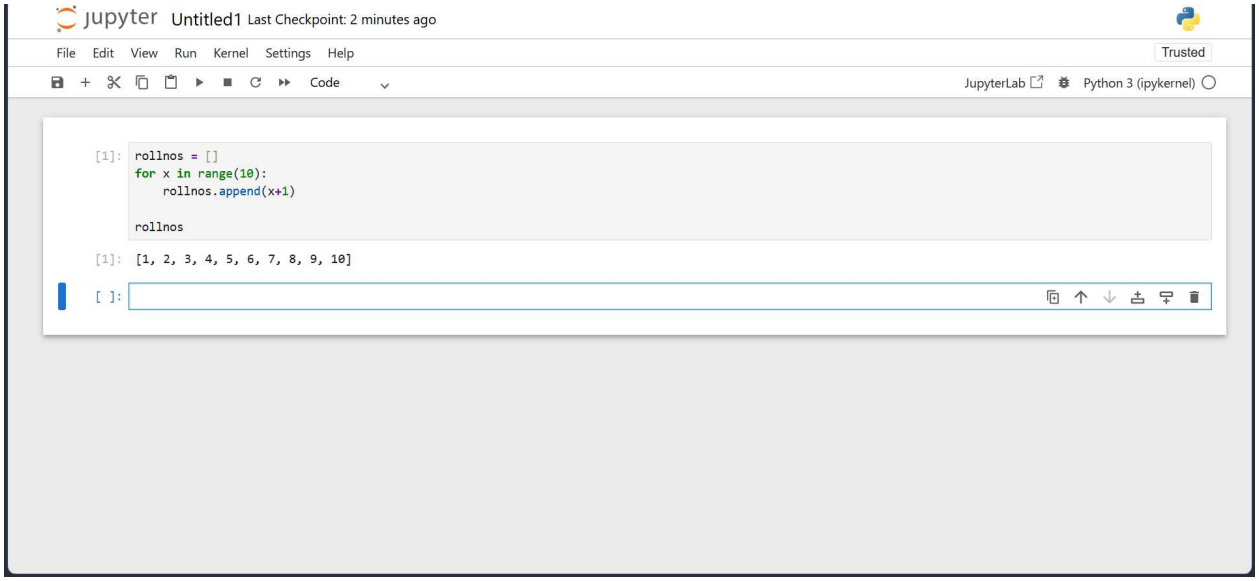
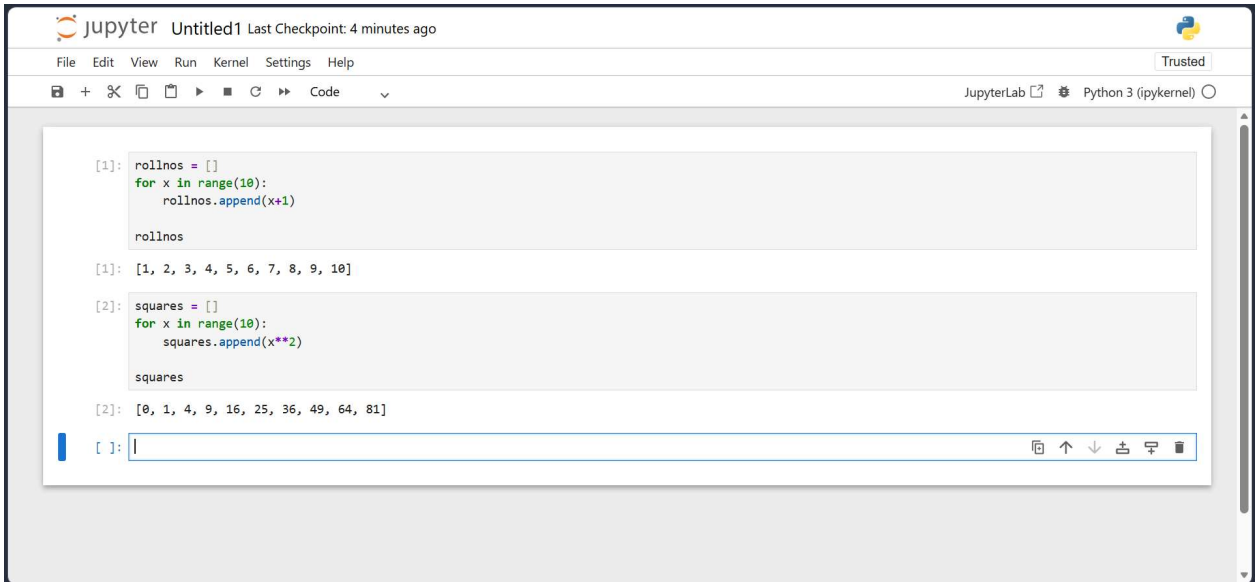
List comprehensions






```
squares = [x**2 for x in range(10)]
```

```
squares
```

```
[0, 1, 4, 9, 16, 25, 36, 49, 64, 81]
```

f-strings(formatted strings)

jupyter Untitled1 Last Checkpoint: 6 minutes ago

File Edit View Run Kernel Settings Help Trusted

Code JupyterLab Python 3 (ipykernel)

```
squares
```

[2]: [0, 1, 4, 9, 16, 25, 36, 49, 64, 81]

```
[3]: squares = [x**2 for x in range(10)]
```

```
[4]: squares
```

[4]: [0, 1, 4, 9, 16, 25, 36, 49, 64, 81]

```
[5]: item = "Laptop"
cost = 70000
print("The item name is ", item, " and cost is ", cost)
```

```
The item name is  Laptop  and cost is  70000
```

```
[6]: item = "Laptop"
cost = 70000
print(f"The item name is {item} and cost is {cost}")
```

```
The item name is Laptop and cost is 70000
```

[ ]:

Multiple assignments:-

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Code JupyterLab Python 3 (ipykernel)

```
*[8]: a, b, c = 3, 4, 5
s = 0
s = a + b + C
s
```

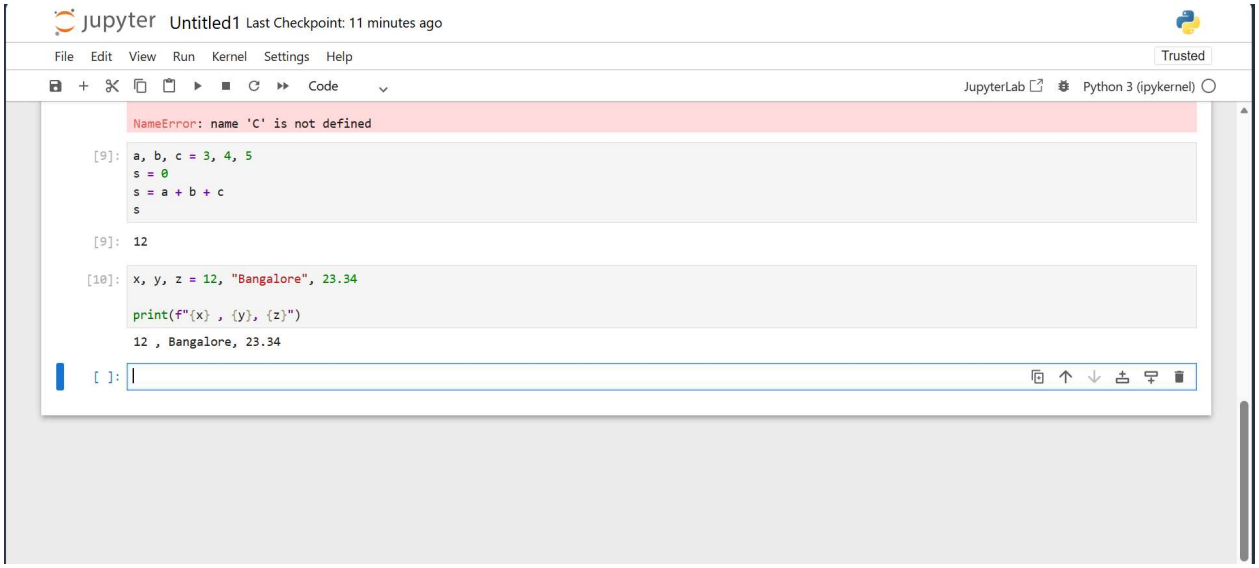
```
---------------------------------------------------------------------------
NameError                                 Traceback (most recent call last)
Cell In[8], line 5
      1 a, b, c = 3, 4, 5
      3 s = 0
----> 5 s = a + b + C
      7 s

NameError: name 'C' is not defined
```

```
[9]: a, b, c = 3, 4, 5
s = 0
s = a + b + c
s
```

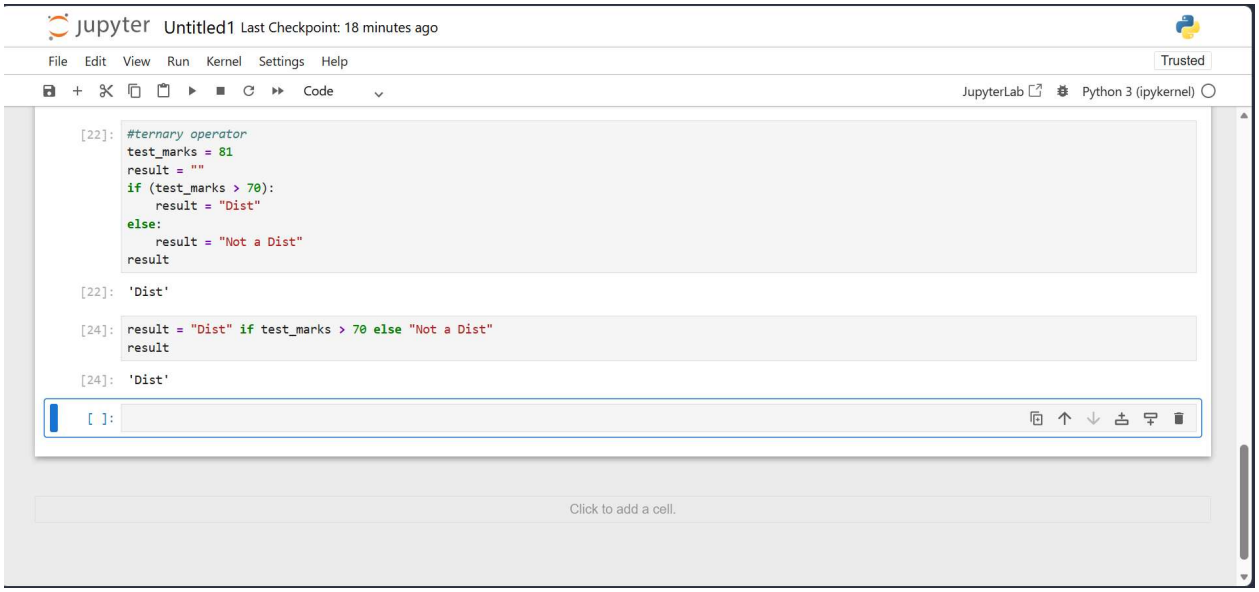
[9]: 12

[ ]:

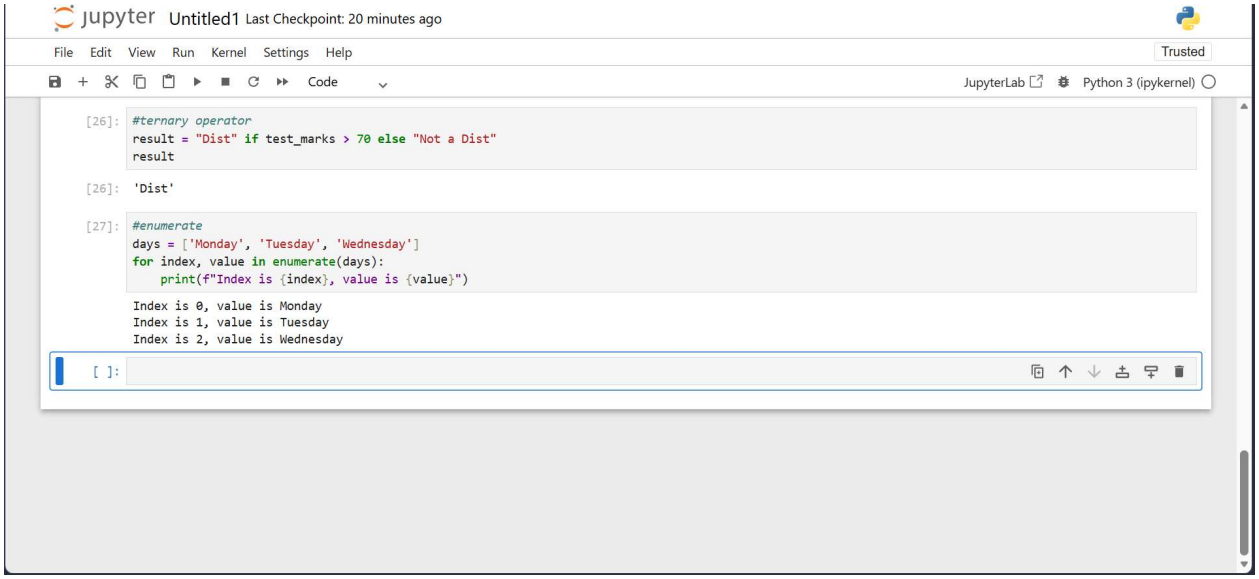


Unpacking:-

Ternary operator:-



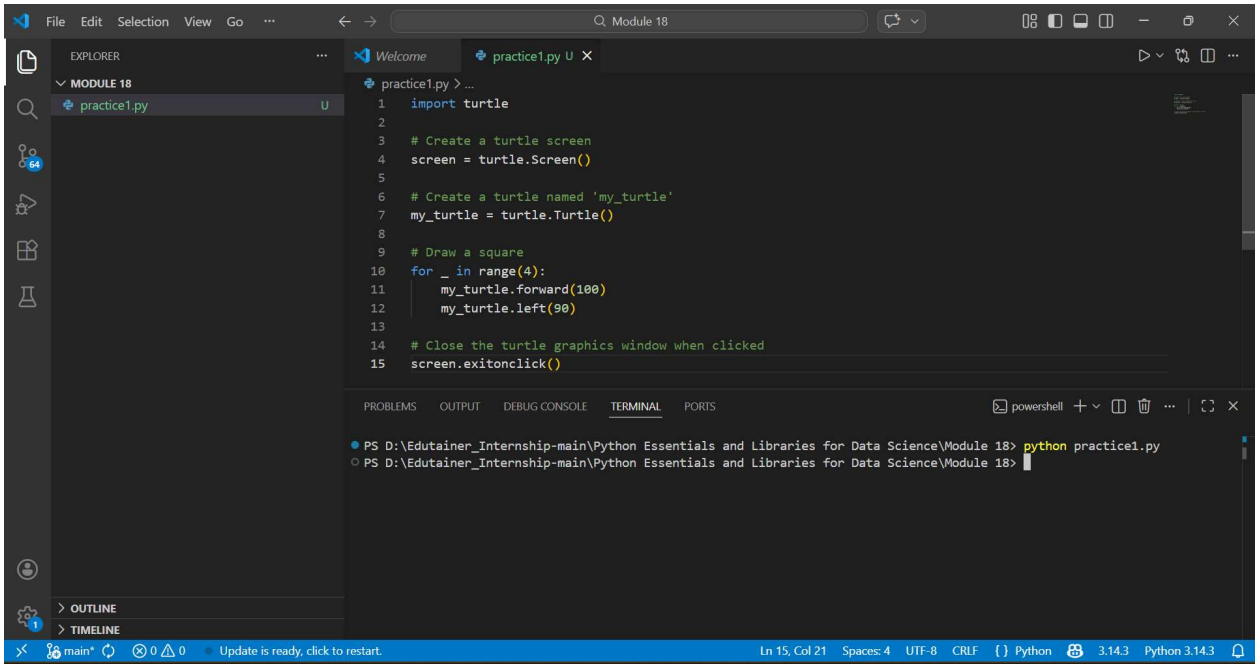
Enumerate:-

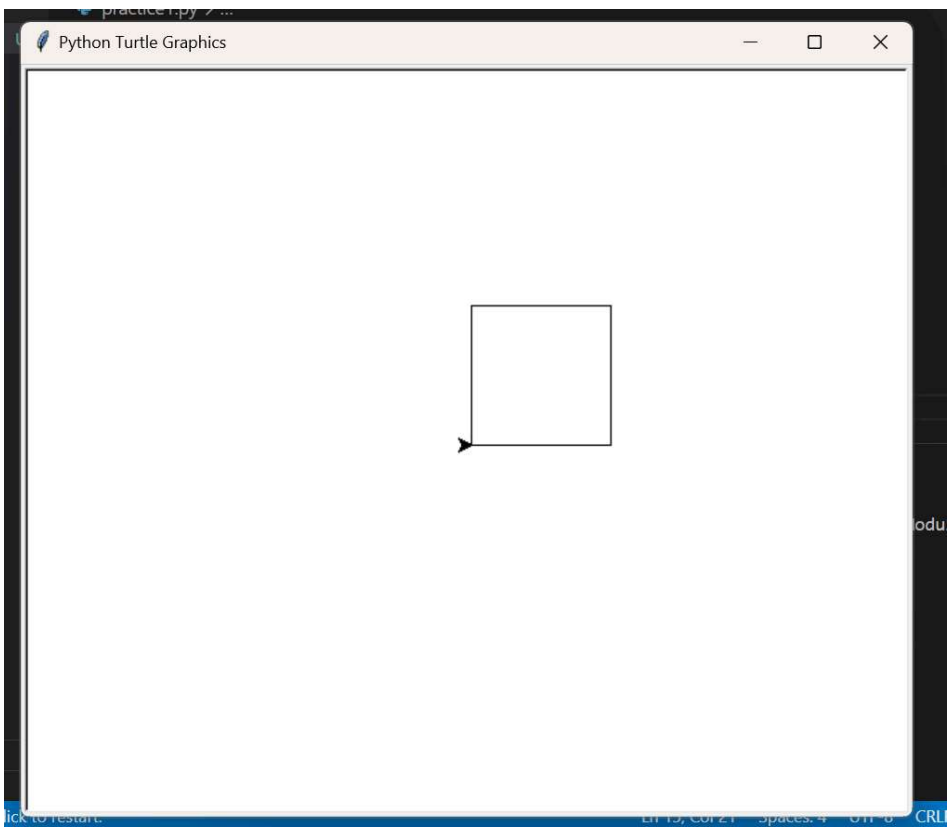


Turtle:-

Python – just for FUN

Let us see how we can create a nice looking graphics output using python module called : **turtle**


File Edit Selection View Go
Module 18
EXPLORER
MODULE 18
practice1.py
Welcome
practice1.py U
practice1.py > ...
import turtle
# Create a turtle screen
screen = turtle.Screen()
# Create a turtle named 'my_turtle'
my_turtle = turtle.Turtle()
# Draw a square
for _ in range(4):
my_turtle.forward(100)
my_turtle.left(90)
# Close the turtle graphics window when clicked
screen.exitonclick()
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
powershell
PS D:\Edutainer_Internship-main\Python Essentials and Libraries for Data Science\Module 18> python practice1.py
PS D:\Edutainer_Internship-main\Python Essentials and Libraries for Data Science\Module 18>
OUTLINE
TIMELINE
main*
Update is ready, click to restart.
Ln 15, Col 21 Spaces: 4 UTF-8 CRLF Python 3.14.3 Python 3.14.3


Python Turtle Graphics

Project:-

Create a simple to-do list application.
• Users can add tasks, view the current tasks, mark tasks as completed, and exit the application.
• The tasks are stored in a list, and the program provides a menu-driven interface for the user to interact with the to-do list.
• What you can learn from this project:

▪ managing a list,
▪ handling user input for list manipulation, and
▪ implementing a menu-driven interface

```
tasks = []



while True:

    print("\n--- To-Do List Menu ---")

    print("1. Add Task")

    print("2. View Tasks")

    print("3. Mark Task as Completed")

    print("4. Exit")



    choice = input("Enter your choice: ")



    if choice == '1':

        task = input("Enter the task: ")

        tasks.append(task)

        print("Task added successfully.")



    elif choice == '2':

        if len(tasks) == 0:
```

```
            print("No tasks available.")
        else:
            print("\nCurrent Tasks:")
            for i, task in enumerate(tasks, start=1):
                print(f"{i}. {task}")

    elif choice == '3':
        if len(tasks) == 0:
            print("No tasks to mark as completed.")
        else:
            print("\nCurrent Tasks:")
            for i, task in enumerate(tasks, start=1):
                print(f"{i}. {task}")

            task_num = int(input("Enter task number to mark as completed:
"))

            if 1 <= task_num <= len(tasks):
                completed_task = tasks.pop(task_num - 1)
                print(f"Task '{completed_task}' marked as completed.")
            else:
                print("Invalid task number.")

    elif choice == '4':
```

```
        print("Exiting the application.")
        break


    else:
        print("Invalid choice. Please try again.")
```

Output

PS D:\Edutainer_Internship-main\Python Essentials and Libraries for Data Science\Module 18> python to-do-list.py

--- To-Do List Menu ---

1. Add Task

2. View Tasks

3. Mark Task as Completed

4. Exit

Enter your choice: 1

Enter the task: eating

Task added successfully.

--- To-Do List Menu ---

1. Add Task

2. View Tasks

3. Mark Task as Completed

4. Exit

Enter your choice: 1

Enter the task: dancing

Task added successfully.

--- To-Do List Menu ---

1. Add Task

2. View Tasks

3. Mark Task as Completed

4. Exit

Enter your choice: 1

Enter the task: sleeping

Task added successfully.

--- To-Do List Menu ---

1. Add Task

2. View Tasks

3. Mark Task as Completed

4. Exit

Enter your choice: 2

Current Tasks:

1. eating

2. dancing

3. sleeping

```
--- To-Do List Menu ---

1. Add Task

2. View Tasks

3. Mark Task as Completed

4. Exit

Enter your choice: 3


Current Tasks:

1. eating

2. dancing

3. sleeping

Enter task number to mark as completed: 1

Task 'eating' marked as completed.


--- To-Do List Menu ---

1. Add Task

2. View Tasks

3. Mark Task as Completed

4. Exit

Enter your choice: 4

Exiting the application.

PS D:\Edutainer_Internship-main\Python Essentials and Libraries for Data Science\Module 18>
```